

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 669 Computational Optimization — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Boyd & Vandenberghe — Convex Optimization (free book) and Stanford EE364A (free lectures)	The primary source. Free book, free video lectures, free problem sets with solutions. Modules 02 through 07 follow its development closely, and the modelling chapter informs Module 12.
MIT 15.093 — Optimization Methods (free, OCW)	Stronger on linear and integer programming than Boyd, which is convex-first. The reference for Modules 03, 09, and 11.
Nocedal & Wright — Numerical Optimization	The reference for the algorithmic detail of Modules 05 and 07 — line searches, trust regions, quasi-Newton. Library copy.
Williamson & Shmoys — The Design of Approximation Algorithms (free PDF)	Free in full, and the reference for Module 10. The LP rounding and randomised rounding chapters are the relevant ones.
CVXPY documentation and examples (free)	Disciplined convex programming — a modelling language that refuses to accept a problem it cannot verify is convex, which is itself instructive.
Bixby — A Brief History of Linear Programming Computation (free)	The Module 13 material: solvers improved by roughly as much from algorithms as from hardware, over the same period.

Module-by-module map

Module 01 — The Shape of the Problem

Resource	What it covers for this module
Boyd & Vandenberghe — Convex Optimization, chapter 1 (free)	The §1 and §2 framing , and the clearest available statement of why convexity rather than linearity is the boundary.
Stanford EE364A — lecture 1 (free video)	Boyd lecturing the same material. Worth watching alongside the chapter.
MIT 15.093 — introduction and problem classification (free)	The §2 hierarchy with more emphasis on the linear and integer end.
Bixby — Solving Real-World Linear Programs: A Decade and More of Progress (free)	Evidence for the §2 callout — what solvers actually achieve on real instances, and how far that is from the worst case.

Module 02 — Convexity

Resource	What it covers for this module
Boyd & Vandenberghe — Convex Optimization, chapters 2 and 3 (free)	The definitive treatment of §1 through §3 , including the full atom library and every preservation rule with proof.
Grant, Boyd & Ye — Disciplined Convex Programming (free)	The §2 callout's ruleset , stated formally — which explains exactly why CVXPY rejects what it rejects.
CVXPY — the atomic function library and DCP tutorial (free)	The atoms of §3 with their curvature and monotonicity, machine-readable and worth browsing.
Boyd — EE364A lectures 2 and 3 (free video)	The same material lectured, with the geometric intuition that the book states more tersely.

Module 03 — Linear Programming

Resource	What it covers for this module
MIT 15.093 — linear programming lectures (free, OCW)	The reference for §1 through §3 , including the geometry and the simplex development.
Spielman & Teng — Smoothed Analysis of Algorithms (free)	The §2 resolution . Read the introduction for the argument even if the analysis is heavy.
Schrijver — Theory of Linear and Integer Programming	The reference for total unimodularity (§3) and the polyhedral theory. Library copy.
HiGHS, and the CVXPY linear programming examples (free)	A free solver that is genuinely competitive, with documentation on the §4 practical concerns.

Module 04 — Duality

Resource	What it covers for this module
Boyd & Vandenberghe — Convex Optimization, chapter 5 (free)	The definitive treatment of this entire module , including the sensitivity interpretation and Slater's condition.
Boyd — EE364A lectures on duality (free video)	The geometric intuition for weak and strong duality, which the algebra alone does not convey.
MIT 15.093 — LP duality and sensitivity analysis (free)	The §2 shadow price material , with worked economic interpretations.
Vanderbei — Linear Programming: Foundations and Extensions (free PDF)	Free, and strong on duality's practical use and on the game theory connection in §3.

Module 05 — First-Order Methods

Resource	What it covers for this module
Nesterov — Introductory Lectures on Convex Optimization	The rates of §1 and the lower bounds of §2 , from the person who proved most of them. Dense; the first two chapters are the relevant ones.
Boyd — EE364B, subgradient and proximal methods (free)	The §3 material , including the proximal operator catalogue and the soft-thresholding derivation.
Beck & Teboulle — FISTA (free)	Accelerated proximal gradient — §2 and §3 combined, and the standard method for the lasso.
Sebastian Bubeck — Convex Optimization: Algorithms and Complexity (free)	A free, modern, and readable treatment of this whole module , with the lower bounds stated clearly.

Module 06 — Constrained Optimisation

Resource	What it covers for this module
Boyd & Vandenberghe — Convex Optimization, chapters 5 and 10 (free)	KKT and the constrained methods of §1 and §2, with the geometric interpretation.
Boyd et al. — Distributed Optimization and Statistical Learning via ADMM (free)	The §3 monograph . Comprehensive, readable, and full of worked formulations including the lasso.
Nocedal & Wright — Numerical Optimization, chapters 12 and 17	The augmented Lagrangian development of §2, with the conditioning analysis.
Absil, Mahony & Sepulchre — Optimization Algorithms on Matrix Manifolds (free PDF)	The manifold case of §4, free from the publisher.

Module 07 — Newton and Interior Point

Resource	What it covers for this module
Boyd & Vandenberghe — Convex Optimization, chapters 9, 10 and 11 (free)	Newton, equality-constrained Newton, and the barrier method with the m/t result — the core of this module.
Nocedal & Wright — Numerical Optimization, chapters 6, 7 and 19	Quasi-Newton methods and the practical detail of §2, including L-BFGS's two-loop recursion.
Martens — Deep Learning via Hessian-Free Optimization (free)	The §2 callout applied to neural networks — Hessian-vector products via automatic differentiation, at scale.
Wright — Primal-Dual Interior-Point Methods	The reference for §3's method as actually implemented in solvers.

Module 08 — Stochastic and Large-Scale Methods

Resource	What it covers for this module
Bottou, Curtis & Nocedal — Optimization Methods for Large-Scale Machine Learning (free)	The reference for this whole module. The clearest treatment of the §1 trade and the §2 noise analysis.
Johnson & Zhang — Accelerating Stochastic Gradient Descent using Predictive Variance Reduction (SVRG) (free)	The §3 method, with the control variate argument.
Goyal et al. — Accurate, Large Minibatch SGD (free)	The linear scaling rule of §2 , its warmup requirement, and where it breaks.
Friedman, Hastie & Tibshirani — Regularization Paths for Generalized Linear Models via Coordinate Descent (free)	The §4 method as implemented in glmnet , including the active set and path-following tricks.

Module 09 — Integer Programming

Resource	What it covers for this module
MIT 15.093 — integer programming lectures (free, OCW)	Branch and bound, cutting planes, and formulation strength — the core of §1 and §2.
Wolsey — Integer Programming	The standard reference, strong on formulation and on the convex hull view of §2.
Klotz & Newman — Practical Guidelines for Solving Difficult Mixed Integer Linear Programs (free)	The §4 checklist, from solver developers. The big-M and symmetry advice is here with worked examples.
Bixby — Solving Real-World Linear Programs, and the MIP benchmark histories (free)	The §4 callout's measurements , and current solver benchmarks.

Module 10 — Approximation and Relaxation

Resource	What it covers for this module
Williamson & Shmoys — The Design of Approximation Algorithms (free PDF)	Free in full, and the reference for this entire module. The LP rounding and randomised rounding chapters cover §2 completely.
Goemans & Williamson — Improved Approximation Algorithms for Maximum Cut (free)	The §3 result. The geometric argument is followable and worth working through.
Vazirani — Approximation Algorithms (free PDF)	Free, and complementary — stronger on the primal-dual method and on the combinatorial algorithms.
Khot — On the Unique Games Conjecture (free survey)	The §3 and §4 conditionality , explained by the person who proposed the conjecture.

Module 11 — Combinatorial Optimisation

Resource	What it covers for this module
Williamson & Shmoys, and Vazirani — the combinatorial chapters (free)	Flow, matching, and the primal-dual method, with the duality connections of §1 made explicit.
Schrijver — Combinatorial Optimization: Polyhedra and Efficiency	The comprehensive reference for §1 and §2. The author's shorter lecture notes are free and are the right entry point.
Oxley — Matroid Theory; and Edmonds' greedy papers	The §2 characterisation and its converse. Edmonds' original papers are short and readable.
Krause & Golovin — Submodular Function Maximization (free survey)	The §3 material , with the applications of the callout and the lazy-greedy speedup that makes it practical.

Module 12 — Modelling

Resource	What it covers for this module
Williams — Model Building in Mathematical Programming	The reference for this module , and one of very few books about formulation rather than algorithms. Library copy; it is worth seeking out.
Klotz & Newman — Practical Guidelines (free)	The §3 diagnostics, including irreducible infeasible subsets and elastic formulations.
Boyd & Vandenberghe — Convex Optimization, chapters 4 and 7 (free)	Worked formulations and the multi-objective material of §2, including the Pareto frontier treatment.
Ben-Tal, El Ghaoui & Nemirovski — Robust Optimization	The formal response to §4 — optimising against the worst case within an uncertainty set rather than against a point estimate.

Module 13 — Solvers and Shipping

Resource	What it covers for this module
Mittelmann — Benchmarks for Optimization Software (free)	Independent, current solver benchmarks across every problem class — the honest basis for the §1 table.
CVXPY, JuMP, and Pyomo documentation (free)	The §1 modelling layers, with examples. JuMP's documentation is unusually good on solver interchange.
Google OR-Tools — CP-SAT documentation (free)	The §1 constraint programming option , which is under-known and frequently wins on scheduling.
Klotz & Newman — Practical Guidelines for Solving Difficult Linear Programs (free)	The §2 numerical diagnostics , from people who support solver users for a living.